Grounding or Lexical Overlap?
A Matched-Pair Study of Internals-Based Faithfulness Probes

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Abstract

We provide a confound-controlled methodology and a released "wrong-question" diagnostic for internals-based, reference-free hallucination detection, which treats *context-faithfulness* (is an answer entailed by the context?) as a linear residual-stream direction. In the standard extractive setup, faithfulness is almost perfectly correlated with *lexical overlap* (r=0.99), so an apparent "grounding probe" may be a string-matching detector. Our matched SUPPORT×OVERLAP construction *partially* decouples the two (halving the confound to r=0.50), and the diagnostic fools an off-the-shelf NLI model. Across four instruction-tuned 7–9B families: (i) within the overlap-controlled stratum a support contrast is linearly decodable, but a *no-context ablation* shows it reflects question–answer compatibility rather than context use, and is inseparable from factoid-correctness by construction; (ii) it aligns across families up to a linear map, yet answer-length, overlap, and factuality directions align equally well, a generic matched-stimulus property, not grounding-specific; (iii) the synthetic direction does not transfer to real hallucinations, scoring below NLI, confidence, and lexical overlap on RAGTruth, where only an in-domain probe trained on human labels recovers a usable signal.

1 Introduction

A grounded language model should answer from the *provided context*, not from its parametric memory or from surface cues. Reference-free, internals-based methods operationalize this by reading a linear direction from the residual stream: much prior work decodes a "truth" or "lying" direction (Azaria and Mitchell, 2023; Marks and Tegmark, 2024; Bürger et al., 2024), and more recent work extends this style of analysis to a *contextual* faithfulness direction (O'Neill et al., 2025), an attractive recipe because it needs no external judge and costs roughly one forward pass. We interrogate this recipe *family* rather than any single system: we probe a representative mass-mean direction rather than reproducing a specific observer-model detector, so our negatives bound *this* synthetic axis (§5.4), not stronger internals detectors such as ReDeEP (Sun et al., 2025), which we do not test. Two distinct notions are routinely conflated under this banner: *parametric-factuality* (is a claim true in the model's memory?) and *context-faithfulness* (is a claim entailed by the given context?). Recent work treats both as activation objects and asks how they relate (Wang et al., 2025), but samples the two notions from *different* datasets, so any measured geometry is confounded with topic and lexicon.

We identify a sharper confound that undermines the faithfulness side specifically. In the standard extractive QA setup used to build such probes (e.g. entity-swap contexts (Longpre et al., 2021)), an answer is context-faithful *iff* the answer string occurs in the context. Faithfulness is therefore almost perfectly correlated with *lexical overlap* (r=0.99 in our data), and a probe that appears to read "grounding" may simply detect string copying: because faithfulness and overlap nearly coincide, a near-ceiling probe *could* be reading either, and telling them apart is what our matched construction (§3) is built to do.

A matched control. We introduce a SUPPORT×OVERLAP construction (§3) that crosses semantic support with lexical presence in a balanced 2×2. Two cells are model-free: a *wrong-question trap*, where the answer is present but answers a different question (supported=no, overlap=yes), and the usual unsupported-absent cell; a paraphrase cell closes the design. This drops $\text{corr}(\text{SUPPORT}, \text{OVERLAP})$ from 0.99 to 0.50, and, tellingly, an off-the-shelf NLI model is fooled by the clean wrong-question trap $\sim$25% of the time, confirming that lexical overlap misleads even "gold" entailment checks.

Contributions. With this control we run a careful study across four model families (Llama-3.1-8B, Qwen2.5-7B-Instruct, Mistral-7B-v0.3, Gemma-2-9B). The result is a confound-controlled methodology and a series of controls that, one by one, attribute the apparent "grounding" signal to factors other than context use.

1. **An overlap-controlled internal signal that reflects question–answer compatibility, not context-grounding (§5.1).** Within the answer-present stratum (lexical overlap held constant), a support contrast is decodable on all four families (collision-free raw AUROC near-ceiling, ≈ 0.99; 0.76–0.88 after purging model confidence). But a *no-context ablation* shows it is mostly question–answer compatibility: removing the context barely changes it (raw 0.95–0.98, confidence-purged unchanged), so the probe separates the matched answer from the wrong-question answer *without reading the context*. Together with the fact that support coincides with our *synthetic* factuality label here by construction, this attributes the overlap-controlled contrast to question–answer compatibility and factoid-correctness rather than context-grounding.

2. **Cross-family alignment up to a linear map (§5.3).** An orthogonal-Procrustes map carries the support direction between families so the transported direction is nearly collinear with the target family's *native* direction (cosine ≈ 0.98, vs. ≈ 0.01 for a random direction through the same map), whereas an anchor-projection transfer reaches only 0.67 AUROC (Kim and Han, 2026). The transferred-probe AUROC (0.91) equals the within-family bound but is ceiling-saturated, and signed map-aware nulls sit at chance (0.52); we therefore rest the claim on the cosine. Factuality, answer-length and overlap directions transport identically while a confidence direction does not, indicating a shared stimulus-derived contrast rather than a grounding-specific axis; we do not generalize beyond these four 7–9B models (Puri et al., 2025).

3. **The confidence asymmetry runs the wrong way (§5.2).** Contrary to a natural hypothesis, confidence absorbs the *support* axis somewhat more than the factuality axis (mean -0.095 over 71 layer-tests), so confidence-purging does not separate the two notions.

4. **Synthetic axes do not transfer to real hallucinations (§5.4).** Applied to RAGTruth (Niu et al., 2024), the controlled-stimulus support direction scores *below* NLI, confidence, and lexical overlap among single-pass API-free baselines; within a task no internals- or NLI-based signal, including a sentence-level SummaC-style NLI baseline, exceeds ≈ 0.6 AUROC, and even lexical overlap, the strongest baseline, reaches only macro 0.647. An in-domain probe trained on RAGTruth human labels recovers a usable signal (0.77–0.81 on the three RAGTruth families; exceeds a conservative single-hypothesis NLI baseline but ties lexical overlap on Qwen), so a real support signal exists in the internals but is a *different direction* from the synthetic axis and needs in-domain annotation.

The takeaway for the "faithful and efficient" agenda is twofold: internals-based support contrasts identified on controlled data are real and remarkably portable across model families (though not grounding-specific), but are not, by themselves, deployable hallucination detectors: on real data, lexical overlap still dominates.

2 Related Work

Geometry of truth and faithfulness. Linear "truth" directions are decodable from activations (Azaria and Mitchell, 2023; Marks and Tegmark, 2024) and partly shared across models (Bürger et al., 2024), but are task-dependent and often fail to transfer across tasks (Azizian et al., 2025; Bao et al., 2025). Wang et al. (2025) treat factuality and faithfulness hallucinations as activation subspaces and argue they overlap; crucially they draw the two notions from different corpora, leaving the geometry confounded with surface form. Adarsh et al. (2026) study how a single truth vector rotates when context is added. We instead hold the two notions on *matched* stimuli and control lexical overlap explicitly; we note, however, that our factuality label is a synthetic gold-answer relabeling rather than a parametric memory-truth probe, so we engage this line methodologically without claiming to adjudicate whether factuality and faithfulness share a direction.

Knowledge conflict and source attribution. Probes can detect knowledge conflict and predict which source a model follows (Zhao et al., 2024; Sun et al., 2025; Brink et al., 2026), and benchmarks such as FaithEval (Ming et al., 2025) score context-versus-parametric faithfulness behaviorally. These are behavioral/source signals; we ask whether a *content* grounding axis exists that is not reducible to lexical overlap.

Cross-model transfer. Linear concept directions can be aligned across models via orthogonal Procrustes (Schönemann, 1966; Puri et al., 2025) or anchor projection (Kim and Han, 2026). We apply both to a support direction; Procrustes renders it nearly collinear with the target's native direction (cosine ≈ 0.98), whereas anchor projection transfers markedly worse in AUROC (§5.3).

Reference-free hallucination detection. Reference-free detectors fall into three families: *external* entailment/alignment scorers (Zha et al., 2023); *internals*-based probes (single residual directions (O'Neill et al., 2025) and the truth/faithfulness directions above); and *confidence*/correctness signals (Cho et al., 2026; Moreno Cencerrado et al., 2025), with faithfulness-aware UQ (Fadeeva et al., 2026) and self-aware RAG abstention (Zhu et al., 2026) combining several. We use one baseline from each family on RAGTruth (external NLI, model confidence, and lexical overlap), and find our controlled-stimulus axis does not beat them.

3 The SUPPORT×OVERLAP Construction

We build matched statements from NQ-Swap (Longpre et al., 2021), which provides, per question q, a gold answer a_{true} supported by an original context c_{org} and a substituted entity a_{swap} supported by an entity-swapped context c_{sub}. We define two bits per (context, answer): SUPPORT S (does the given context entail the asserted answer for the asked question?) and OVERLAP O (does the answer string occur in the context?). The 2×2 design has four cells, of which we realize three (cell B is unrealized in this release; see below):

cell	(question, context, answer)	S	O
A	$(q,\ c_{\text{org}},\ a_{\text{true}})$	1	1
B	$(q,\ \text{paraphrase}(c_{\text{org}}),\ a_{\text{true}})$	1	0
C	$(q',\ c_{\text{org}},\ a_{\text{true}})$	0	1
D	$(q,\ c_{\text{sub}},\ a_{\text{true}})$	0	0

Cell C is a *wrong-question trap*: a_{true} is present in c_{org} but is offered as the answer to a different question q' (the next item's question). Cell B would paraphrase c_{org} so the answer is entailed but not copied (open instruct model, keeping an item only if token-overlap≤ 0.34 and an NLI model still entails it); the released build supplies no paraphraser, so cell B has *zero* yield and all reported analyses use the model-free cells A/C/D with $\text{corr}(S, O)=0.50$ (0.56 after removing the donor-collision items below; an orthogonalized $\text{corr}\approx 0$ design would require cell B). One caveat in cell C: NQ-Swap dev has only 418 distinct questions over 1,200 items, so for 278 items (23.2%) the donor q' is string-identical to q and a_{true} *does* answer it; we flag these *donor-collision* items and report cell-C results with them removed (the full construction will dedup $q' \neq q$). On *clean* cell C an off-the-shelf NLI checkpoint (Laurer et al., 2024; He et al., 2023) still rates 25.5% of unsupported-but-present items as "supported" (mean entailment 0.31, vs. 0.74 for cell A and 0.18 for cell D); the raw 37.6% rate is inflated by the collisions, which NLI entails 77.7% of the time (correctly). Lexical overlap thus misleads "gold" entailment even after the label error is purged (Fig. 1).

4 Probing, Separability, and Transfer

Directions. For a (model, layer) we extract last-token residual states X over the asserted answer span (one forward pass, bf16, left-padded). A SUPPORT direction d_S and FACTUALITY direction d_F are mass-mean (difference-of-means) estimates (Marks and Tegmark, 2024); signed projections Xd are the probe scores. Every reported number uses one protocol. Folds are grouped by *source*, not by record: by the normalized (question, context, answer) triple of the underlying NQ-Swap QA for CtrlPairs (1,200 records, 555 triples) and by source passage for RAGTruth (2,000 responses, 450 passages), since ITEM_ID indexes one swap record or one response and does not isolate sources. Residualization is fitted *within* each training fold. AUROC is *signed*: polarity is fixed on the training fold and the test projection scored as-is, with no $\max(\text{AUC}, 1-\text{AUC})$ flip anywhere, so a non-predictive direction averages to 0.5 instead of being floored above it. Folds come from a seeded group splitter; estimates are mean±sd over ten splits with source-cluster bootstrap CIs.

Overlap control. To test grounding *beyond* overlap we (i) restrict to the answer-present stratum $O=1$ (cells A vs. C), where lexical presence is con-

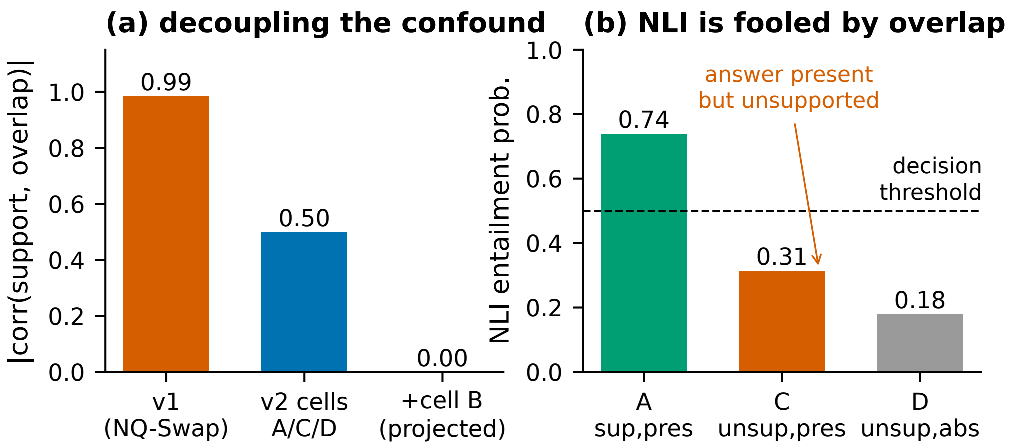

Figure 1: **Partially decoupling support from lexical overlap.** (a) The matched construction drops $|\mathrm{corr}(\textsc{support}, \textsc{overlap})|$ from 0.99 (extractive NQ-Swap) to 0.50 (model-free cells A/C/D), and would reach ≈ 0 with the paraphrase cell B (projected; cell B has zero yield in this release). (b) Mean NLI entailment by *clean* cell: on the wrong-question trap (C), where the answer is present but unsupported, NLI is still pulled toward "supported."

stant, and (ii) residualize X on surface covariates (overlap, length) before re-probing.

Null-calibrated separability. We treat each notion as an undirected axis and measure the acute angle $\angle(d_S, d_F)$. Significance is assessed against a within-notion null built by splitting one notion's data in half, re-estimating the same axis on each half and measuring the half-vs-half angle; this is a resampling null, not a label permutation. With n_{null}=2,000 draws per notion we report the resolution bound $p<5\times10^{-4}$ rather than p=0 when no null draw exceeds the observed angle. It is run per family at the a-priori mid layer on all three cells; it cannot be posed *within* the overlap-controlled stratum, where S and F coincide exactly.

Confidence purge. We regress each hidden dimension on three confidence covariates (mean log-prob, mean max-softmax, first max-softmax) and re-probe the residual, reporting the fraction of probe signal absorbed per axis.

Cross-family transfer. Families differ in hidden size, so we fit a semi-orthogonal Procrustes map between two models' residual spaces on a shared calibration split, transport a source-trained probe, and measure target AUROC. The calibration/evaluation split is a *GroupKFold on* ITEM_ID (items in the calibration set are disjoint from the evaluation set), so the map is never fit and tested on near-duplicate rows of the same item; transported-probe polarity is fixed on the target calibration fold. We compare to the anchor-projection ACS baseline (Kim and Han, 2026), a within-family upper bound, and a random-direction floor (also scored without the polarity flip, so it sits at chance rather than being inflated above 0.5).

5 Experiments

Setup. Four instruction-tuned families: Llama-3.1-8B (Llama Team, 2024), Qwen2.5-7B (Qwen Team, 2024), Mistral-7B-v0.3 (Jiang et al., 2023), Gemma-2-9B (Gemma Team, 2024). CtrlPairs: 1,200 NQ-Swap items $\rightarrow$ 3,593 matched statements (cells A/C/D). RAGTruth: 2,000 test responses with response-level hallucination labels (Niu et al., 2024). Baselines: NLI (the `DeBERTa-v3-base-mnli-fever-anli` checkpoint (Laurer et al., 2024), DeBERTaV3 (He et al., 2023) fine-tuned on MNLI, FEVER and ANLI (Williams et al., 2018; Thorne et al., 2018; Nie et al., 2020)) in two variants, (i) *whole-response*: a single (context, declarativized-answer) entailment query (deliberately simple and conservative for multi-sentence responses); and (ii) *sentence-level*, a SummaC-style decomposition (Laban et al., 2022) that entailment-checks each response sentence against every context sentence, takes the max over context, and aggregates over response sentences by mean (SummaC-ZS) or min (for tractability we cap context at 60 sentences and pairs at 256 tokens, which can only lower SummaC and is thus conservative), plus max-softmax confidence and lexical overlap. Everything is API-free; the NLI baselines use the small DeBERTa model and all activation extraction runs on one A100-40G.

5.1 The within-stratum signal is question–answer compatibility, not grounding

Leakage control. Grouping on ITEM_ID is not sufficient: it isolates swap records, not source QAs, and 903 of the 1,200 cell-A rows have a byte-identical twin under a different ITEM_ID (mid-layer cosine 0.9999). We group by the source QA triple instead (555 groups), for the C1 cross-validation, the Procrustes calibration split (§5.3) and the cross-strata control, which is additionally evaluated on *held-out* groups. Regrouping moves every C1 estimate by at most 0.007 (App. A): the leak is real but inert, since a difference of class means in $\sim$3500 dimensions barely shifts under a few duplicated rows. To avoid selecting the layer on the very number we report, we fix the analysis layer *a priori* at relative depth 0.5 (the nearest cached layer to $0.5\,L_{\text{tot}}$: L=14/16/16/21 for Qwen/Mistral/Llama/Gemma), matching the relative-depth convention already used for C3/C4 and for Fig. 2; App. A reports the mean±sd across the mid-layer band (0.4–0.6 depth), showing the result is not a single-layer artifact.

Table 1 reports support decodability within the answer-present stratum O=1 (cells A vs. C), where lexical overlap is held constant, on the *collision-free* cells: we drop the 23.2% of cell-C items whose donor question is string-identical to the original (§3), which would otherwise mislabel genuinely-supported items as unsupported. Three observations bound the interpretation. First, the raw within-stratum AUROC is *near-ceiling* (0.99; 0.988–0.999 across families): once mislabels are removed, A and C are almost perfectly separable. We read this less as strong grounding than as evidence that the contrast is *easy*: within O=1 it varies *only the question* (context and answer are fixed), so a probe can in principle separate A from C by question–answer appropriateness without using the context. Second, the operative confound is *model confidence* (overlap and answer length are uninformative here, AUROC 0.500): answer log-prob and NLI separate A from C on their own, so we residualize the hidden states on log-prob and max-softmax; the support contrast then falls to 0.76–0.88, which we take as the primary confound-controlled estimate. Third, and decisively, a *no-context ablation* settles what the signal is: re-extracting the *A*/*C* activations with the context *removed* (question+answer only) leaves $\text{supp}_{O=1}$ almost unchanged (raw 0.95–0.98 vs. 0.99 with context; the confidence-purged estimate does not collapse either, 0.80–0.83, rising for two families and falling only $\sim$0.08 for Mistral), and replacing the context with an unrelated passage gives the same (0.94–0.98). The within-O=1 contrast is therefore mostly *question–answer compatibility*: the probe separates A (whose question the answer fits) from C (whose donor question it does not) *without reading the context at all*. A cross-strata control (train A vs. D, which fixes the question but also varies overlap since D is answer-absent; evaluate A vs. C) does transfer at 0.66–0.88, so *some* context-sensitive component exists, but it is an upper bound and not separable from the dominant compatibility signal here. Finally, support and our *synthetic* factuality label are *identical* within O=1 by construction (they diverge only in the answer-absent cell D). We therefore do *not* claim a context-grounding signal: within the overlap-controlled stratum the decodable contrast is question–answer compatibility, not isolated from factoid-correctness, and not shown to use the context at all.

model	L	$\text{supp}_{O=1}$	+conf	cross
Qwen2.5	14	0.988	0.760	0.661
Mistral	16	0.996	0.878	0.870
Llama	16	0.997	0.789	0.876
Gemma	21	0.998	0.811	0.827

Table 1: **C1 (confound-controlled, collision-free).** Layer fixed a priori at relative depth 0.5; folds grouped by *source QA* (555 groups over 1,200 records); polarity resolved on the train fold and scored signed; donor-collision cell-C mislabels removed (§3; n_A=1200, n_C=922). Each entry is the mean over ten seeded splits (sd $\leq$ 0.002). $\text{supp}_{O=1}$ is support AUROC within the answer-present stratum, near-ceiling once mislabels are removed, because the contrast varies only the question (source-cluster 95% CIs [0.983, 0.992], [0.993, 0.998], [0.995, 0.999], [0.997, 0.999]). *+conf*: hidden states residualized on the three confidence scalars, fitted within each fold; our primary estimate. *cross*: train A vs. D, evaluate A vs. C on *held-out* groups. Within O=1 support and the synthetic factuality label coincide, so we make no distinctness claim.

5.2 The confidence asymmetry runs the wrong way

On the full three-cell sample the two axes are not collinear: at the a-priori layer $\angle(d_S, d_F)$ is

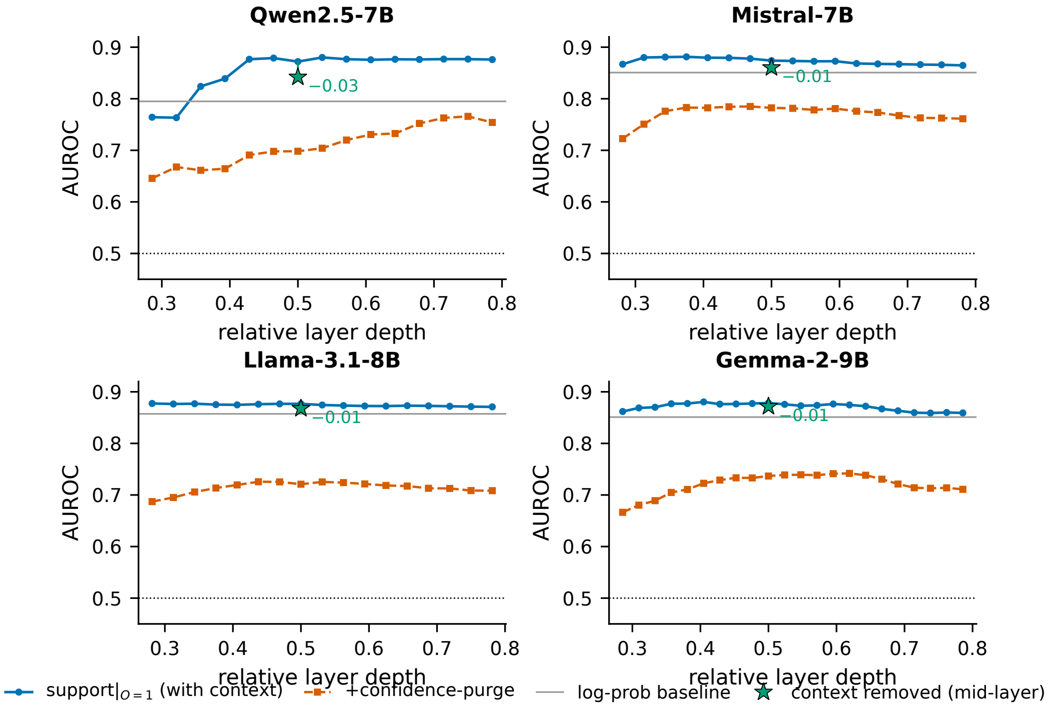

Figure 2: **The within-stratum contrast peaks mid-layer, but it is not grounding.** Cross-validated AUROC vs. relative layer depth for the support contrast within the answer-present stratum (O=1, where overlap is uninformative): raw, and after purging model confidence; the grey line is a log-probability baseline. The contrast is high and stable through mid layers on all four families. Curves use the same source-grouped, signed protocol as Table 1 but *retain* the donor-collision mislabels, hence their lower level (≈ 0.87 raw): removing those items is what lifts the contrast to near-ceiling. The green star marks the same probe at the a-priori layer with the context *removed*: the raw AUROC barely moves (drop annotated), and shuffled context behaves the same. The purged value even *rises* for Qwen (0.757→0.827) and Llama, is flat for Gemma, and falls materially only for Mistral (0.878→0.802). The contrast survives removing the context, so it is mostly question–answer compatibility, not context use.

78°/54°/67°/65° for Qwen/Mistral/Llama/Gemma, each beyond every draw of its within-notion null ($p<5\times10^{-4}$). This says little, though: those stimuli have $\mathrm{corr}(S,F)$=0.50 by construction, and inside the overlap-controlled stratum the labels coincide exactly, so the angle cannot establish an overlap-independent axis.

We also hypothesized that confidence absorbs factuality more than faithfulness, which would separate them. It does not. Over every cached layer of all four families (71 tests), $\mathrm{absorbed}(F)-\mathrm{absorbed}(S)$ averages -0.095 (sd 0.102): 57/71 are negative and 54 are one-sided significant *opposite* to the hypothesis, against 10 favouring factuality (per-family means $+0.002$, -0.083, -0.070, -0.190; Gemma is negative at all 22 layers). Confidence absorbs the support axis somewhat *more*, so purging it is not a route to distinguishing the two notions.

5.3 Matched-stimulus contrasts align across families up to a linear map

We ask whether the support direction occupies the *same* geometry across families up to a linear map. The direct evidence is geometric: the Procrustes-mapped source direction is nearly collinear with the target family's *native* direction (transported-to-native cosine 0.976, range 0.95–0.99 over the 12 ordered pairs; Table 2); anchor projection reaches only 0.67 transfer AUROC against Procrustes' 0.906 (the two are not comparable in cosine, as ACS has no target-space direction). The cosine is far above its null: a random or label-shuffled source direction transported through the *same* fitted map is essentially orthogonal to the native direction (cosine 0.01 and 0.10 respectively), so 0.976 is not an artifact of the shared-stimulus geometry. It is also not an artifact of fitting the map and the directions on the same items: with the map fit on a calibration split and the target's *native* direction re-estimated on a disjoint, item-grouped held-out split, the

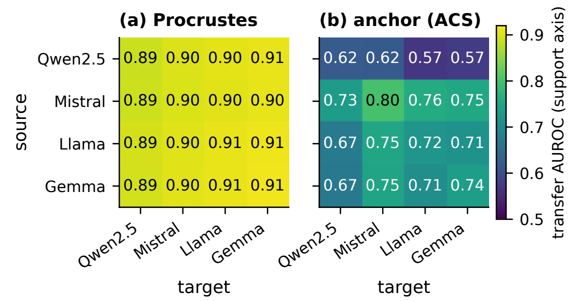

Figure 3: **Matched-stimulus contrasts align across families up to a linear map.** Source→target transfer AUROC of the support direction; every cell, diagonal included, uses that panel's own estimator, source-grouped. (a) Orthogonal Procrustes is uniformly high on- and off-diagonal (0.89–0.91); we read this as alignment (cosine 0.98, Table 2), not lossless transfer, since the AUROC is ceiling-saturated. (b) Anchor projection (ACS) is uniformly lower (0.57–0.80) and no better on its own diagonal (0.62–0.80) than across families, so its gap to Procrustes is not a transfer effect.

transported-to-native cosine is 0.971 ± 0.014 (mean over 12 pairs $\times$ 10 splits; 0.970 with both source and target directions held out from the map), essentially unchanged from the in-sample value (Table 2); the $\sim$0.006 drop matches the native direction's own cross-split self-consistency loss (0.994), so the Procrustes map adds no measurable inflation (factuality behaves identically, 0.990→0.982). We deliberately do *not* headline the transferred-probe AUROC (0.906): it equals the within-family bound (0.905) and is therefore ceiling-saturated; scored with polarity-honest (signed) AUROC, random and shuffled-label directions through the map sit at chance (0.52/0.51). **This alignment is not specific to grounding.** Transporting the *factuality*, *answer-length*, and *lexical-overlap* directions through the same maps gives cosines of 0.99/0.98/0.99, as high as support, whereas a purely model-internal *confidence* direction transports at only 0.50 (and unstably, 0.04–0.90 across pairs). So what aligns is any contrast *derived from the shared input stimuli*, a shared low-rank stimulus geometry, not a grounding-specific axis; that confidence does *not* align shows the alignment is non-trivial but also that it carries no grounding-specific content. We therefore conclude only that *matched-stimulus contrasts, support among them, are linearly alignable across these four* 7–9*B families*, and avoid any claim of a grounding-specific or near-lossless transfer.

axis	cosine	AUROC	within	rand-map	rand-lab
SUPPORT	**0.976**	0.906	0.905	0.519	0.506
FACTUALITY	0.989	0.915	0.911	—	—
nuisance directions:					
length	0.976	—	—	—	—
overlap	0.992	—	—	—	—
confidence	0.504	—	—	—	—

Table 2: **C3 (cosine-based, leakage-controlled).** Mean over the 12 cross-family ordered pairs, GroupKFold on ITEM_ID. *cosine*: transported-to-native cosine (the alignment evidence; non-saturating); shown in-sample, it stays at 0.971 when the native direction is re-estimated on an item-grouped held-out split (§5.3), so it is not a map-fitting artifact. A random / label-shuffled source direction through the *same* map has cosine only 0.01 / 0.10 with the native direction. *AUROC/within*: transferred-probe vs. within-family AUROC: equal, hence ceiling-saturated. *rand-map/rand-lab*: random / shuffled-label directions through the map, signed AUROC, at chance. Support, factuality, answer-length and overlap directions all align at ≈0.98, while a model-internal confidence direction does not (0.50): alignment is a generic property of contrasts derived from the shared stimuli, not a grounding-specific axis.

5.4 Synthetic axes do not transfer to real hallucinations

We learn d_S on CtrlPairs and apply it to the fixed RAGTruth responses (the NLI and overlap baselines are response-intrinsic, hence identical across models), predicting the human faithfulness label (Table 3). Here we keep the *support-polarity* sign fixed from CtrlPairs and score without the test-fold flip, because flipping per-test inflates a near-chance axis; CtrlPairs and RAGTruth share no items, so there is no leakage in this transfer. Scored honestly, the label-free synthetic axis reaches only 0.54–0.66 AUROC (and drops toward chance for the weakest family), *below* NLI (0.673), model confidence (0.62–0.72), and plain lexical overlap (0.753), which is the single best predictor on RAGTruth. Paired item cluster-bootstraps confirm the ordering against overlap (Δ=$-0.097/-0.162/-0.212$) and confidence (all $p \leq 0.015$), though against whole-response NLI only Llama and Gemma separate ($p<0.001$); Qwen ties (Δ=-0.016, CI $[-0.049, +0.015]$). Removing the polarity flip can only lower these numbers, so the negative is if anything understated; on real data overlap still dominates. A more sophisticated sentence-level NLI baseline (SummaC-style, max entailment over context sentences) does no better: pooled 0.614/0.664 and within-task

model	label-free synth-d_S	conf	labelled in-dom[†]	NLI	overlap
Qwen2.5	0.657	0.723	0.765	0.673	0.753
Llama	0.591	0.623	0.814	0.673	0.753
Gemma	0.541	0.686	0.797	0.673	0.753

Table 3: **C4 (leakage-controlled).** RAGTruth hallucination detection (AUROC; three families, Mistral's RAGTruth extraction OOM'd). The label-free synthetic support axis (synth-d_S, support polarity fixed from CtrlPairs, a-priori mid layer, scored without the test-fold flip) is below NLI, confidence, and lexical overlap, and for the weakest family (Gemma, 0.541) is barely above chance. [†]uses RAGTruth human labels (not label-free); source-passage cluster-bootstrap 95% CIs $[0.737, 0.786]$, $[0.780, 0.824]$, $[0.776, 0.820]$.

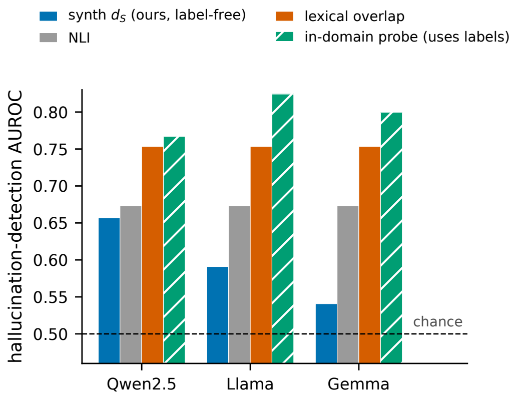

Figure 4: **The synthetic axis does not transfer to real hallucinations.** On RAGTruth, the label-free synthetic support axis (ours) sits below NLI and below plain lexical overlap, the strongest baseline; only an in-domain probe trained on human labels beats NLI. Axis starts at chance (0.5).

macro $0.596/0.579$ (mean/min), essentially tied with the whole-response NLI (0.582) and still well below overlap (Table 4); many RAGTruth hallucinations are abstractive errors that survive sentence-level entailment. An *in-domain* probe (5-fold grouped-CV L_2-logistic on the same mid-layer states, target=human faithfulness label) does recover a usable signal: AUROC $0.77/0.81/0.80$ for Qwen/Llama/Gemma (passage-cluster 95% CIs in Table 3). It exceeds our NLI baseline (0.673) on all three families (a conservative whole-response check, §5); against the strong overlap baseline (0.753) it wins on Llama and Gemma (paired bootstrap $p<0.001$) but only ties on Qwen ($+0.01$, $p=0.17$). Residualizing lexical overlap and answer length out of the features first, within each training fold, leaves 0.68–0.73: some of this signal is overlap, but a substantial internal component is not. A real support signal therefore exists in the internals, but it is a *different direction* from the synthetic axis and is recoverable only with in-domain human annotation.

signal	Data2txt	QA	Summary	macro
synth-d_S	0.487	0.611	0.658	0.585
confidence	0.576	0.544	0.679	0.600
NLI (whole-resp.)	0.547	0.567	0.633	0.582
Sent-NLI (mean)	0.611	0.557	0.620	0.596
Sent-NLI (min)	0.521	0.591	0.626	0.579
overlap	0.605	0.686	0.649	0.647
in-domain[†]	0.766	0.680	0.673	0.706
faithful-rate	0.357	0.823	0.795	—

Table 4: **C4, per task type.** Within-task RAGTruth AUROC; synth-d_S, confidence and in-domain are means over the three models, NLI and overlap are response-intrinsic. Because faithful base rates differ sharply across tasks (bottom row), the pooled numbers in Table 3 overstate every signal — overlap $0.753 \to 0.647$, NLI $0.673 \to 0.582$, the axis $0.596 \to 0.585$. *Sent-NLI* is the SummaC-style sentence-level baseline. Within a task every label-free signal is weak (0.58–0.65 macro; the axis is below chance on Data2txt), and overlap is merely the least-bad. [†]uses RAGTruth human labels (0.706 macro).

6 Conclusion

Our results separate two questions usually merged. *Does* a non-overlap signal exist in the internals? Within the overlap-controlled stratum a contrast is decodable (0.76–0.88 purged) and aligns across four families up to a linear map, but a no-context ablation shows this within-stratum contrast is mostly *question–answer compatibility*, not context use, and the alignment is a generic matched-stimulus property (length and overlap directions align equally well). An apparent "grounding" signal thus survives an overlap control yet is still not grounding. *Is* that direction a deployable, label-free hallucination detector? No, it does not transfer to RAGTruth, where lexical overlap remains the strongest single-pass baseline (and even that edge is largely a between-task base-rate artifact: Table 4). The gap between a clean controlled signal and real-world utility is the main lesson, and it cautions against reading deployment claims from synthetic-stimulus probes; whether any label-free signal can decisively beat lexical overlap on real hallucinations is the natural next question.

Limitations

We are explicit about what we do *not* show. (i) Within the answer-present stratum the decodable contrast is question–answer compatibility (a no-context ablation leaves it almost unchanged, raw 0.95–0.98) and coincides with our synthetic factuality label by construction, so we cannot claim a context-grounding signal there; only a question-fixed cross-strata control (0.66–0.88) shows any context-sensitive component. (ii) Cell B has zero yield, so the fully-orthogonalized ($\mathrm{corr}{\approx}0$) design is not instantiated and all analyses use cells A/C/D at $\mathrm{corr}{=}0.565$ (collision-free) with overlap stratification; the realized data is a three-cell partial factorial, not a balanced 2×2, and within $O{=}1$ the support and synthetic factuality labels are the identical vector, so $\mathrm{supp}_{O=1}$ is simultaneously a factuality AUROC. (iii) The cosine alignment is measured on shared stimuli, so it shows alignment of a shared low-rank contrast, not a fully unpaired transfer. (iv) We evaluate a deterministic strided 2,000-row subset of the 2,700-row `wandb/RAGTruth-processed` test split (450 source passages, 4–5 responses each), covering three task types and one annotation scheme, and the negative is on $3/4$ families (Mistral's RAGTruth extraction ran out of memory at the longer context length and was not retried at a smaller batch size). (v) All probes read the residual stream at the *final answer token*. CtrlPairs answers average 1.7 tokens, so this is near-lossless there; RAGTruth responses average 126.6 tokens, so last-token pooling discards most of the response. We extracted no other pooling and cannot test the alternative here, so the RAGTruth results should be read as specific to this choice. Relatedly, all four instruction-tuned models are probed with a single raw completion template rather than their chat templates; this sensitivity is untested. (vi) Support labels are validated automatically and by a 40-item manual audit of clean cell C ($0/40$ mislabelled), not by a multi-annotator study with agreement statistics. (vii) Stronger internals detectors (e.g. ReDeEP (Sun et al., 2025)) and sampling-based methods are out of our single-A100 budget, so we show that *this* synthetic axis fails, not that internals cannot work.

Ethical Considerations

This paper reports a negative result about a class of proposed detectors, and its main practical risk is over-trust in the opposite direction: a label-free internals probe of the kind we study reaches only 0.54–0.66 AUROC on RAGTruth, below plain lexical overlap (0.753), so deploying one as a hallucination guard would give a false sense of coverage. We therefore recommend that internals-based grounding probes be validated against a lexical-overlap baseline on in-domain human labels before any deployment claim is made. CtrlPairs is constructed automatically from NQ-Swap Wikipedia text and involves no new human subjects, no new annotation, and no personally identifying information beyond what the source corpora already contain; the "wrong-question" items are deliberately mislabelled stimuli for probing and are not intended as training data. All experiments are inference-only on a single A100-40G, with no model training or fine-tuning.

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A Implementation Details and Reproducibility

Code, the constructed datasets and all result JSONs are available at `https://github.com/Beiciccc/GroundLM`; the 2.4 GB of cached residual-stream features that the analysis scripts consume are archived separately at `TODO-ARCHIVE-DOI`, since they exceed what a source repository can hold. With those features in place, `reproduce.sh` regenerates every number, table and figure on CPU; `reproduce.sh extract` regenerates the features themselves on one A100-40G. Below we fix every choice a reader would need to re-run the study.

Layer choice and fold-split sensitivity. Table 1 fixes the analysis layer a priori at relative depth 0.5. Averaging the same collision-free, item-grouped estimate over every cached layer in the 0.4–0.6 band gives $\text{supp}_{O=1}$ of 0.990 ± 0.004 (Qwen, 5 layers), 0.996 ± 0.002 (Mistral, 7), 0.997 ± 0.001 (Llama, 7) and 0.998 ± 0.001 (Gemma, 9), and confidence-purged values of 0.761 ± 0.016, 0.876 ± 0.003, 0.786 ± 0.003 and 0.803 ± 0.009: the result is not a single-layer artifact. We also avoid `GroupKFold`, which assigns folds by sorting group sizes: our group sizes are almost all tied (1193/1200 CtrlPairs items have three rows; every RAGTruth response is its own record), so its split depends on `numpy`'s `argsort` tie-breaking and changes with the library version – enough to move the in-domain probe by 0.009. All reported estimates instead use an explicitly seeded group splitter and are means over ten splits; the split-to-split sd is ≤ 0.002 for Table 1 and ≤ 0.007 for Table 3.

Parametric-belief check. Before probing we verify closed-book that each model actually prefers the gold answer: with the context removed, we compare the mean answer-token log-probability of a_{true} against a_{swap} per item. The gold answer wins for 84.0% (Qwen), 92.6% (Mistral), 94.8% (Llama) and 95.2% (Gemma) of the 1,200 items. All reported results use the full set; the extraction code also supports restricting the analysis to this "knows-gold" subset (`--knows-gold-only`).

Artifacts and licenses. We use NQ-Swap (Longpre et al., 2021) and RAGTruth (Niu et al., 2024) under the MIT license, and the DeBERTaV3 MNLI/FEVER/ANLI checkpoint (He et al., 2023) (MIT). The probed checkpoints are Qwen2.5-7B-Instruct and Mistral-7B-Instruct-v0.3 (Apache-2.0), Llama-3.1-8B-Instruct (Llama 3.1 Community License; this work is "Built with Llama") and Gemma-2-9B-it (Gemma Terms of Use). All use is non-commercial academic research, consistent with the intended use each artifact states. Our code is released under the MIT license; the constructed CtrlPairs data derives from NQ-Swap and inherits its terms. The redistributed RAGTruth subset contains third-party source text that we did not filter for incidental contact details already present upstream.

Prompt and answer span. Activations are extracted from a raw template (no chat formatting): `"{context}\n\nQuestion: {question}\nAnswer:"`, to which the *asserted answer*, a teacher-forced gold surface string (NQ-Swap $a_{\text{true}}/a_{\text{swap}}$ for CtrlPairs; the model-generated response for RAGTruth), is appended with a single leading space and no special tokens. Inputs are *left*-padded, so the answer occupies the final positions. The probe feature $\texttt{last}_L$ is the last-token residual state at layer L (under left padding, the final answer token). This is the *only* pooling we extracted: the released caches contain no answer-span mean, so the sensitivity of our results to pooling is untested (Limitations).

Confidence scalars. Computed over the answer span only (logits shifted by one to predict the answer tokens): *mean log-prob* (mean gold-token log-probability), *mean max-softmax* (mean per-position top-1 probability), and *first max-softmax* (top-1 probability of the first answer token). The confidence confound set in the purge comprises three scalars: mean log-prob, mean max-softmax, and first max-softmax.

Lexical overlap. Two quantities, both case-folded. OVERLAP (continuous, `lexical_overlap`) is the fraction of whitespace-tokenized answer tokens that occur as a substring of the lowercased context; $\text{OVERLAP}_{\text{measured}}$ (binary, the cell label) is whether the *whole* stripped lowercased answer string is a substring of the lowercased context. A cell is dropped when its measured presence disagrees with its intended overlap label.

Construction and the missing cell B. CtrlPairs cells: A= $(q, c_{\text{org}}, a_{\text{true}})$; C= $(q', c_{\text{org}}, a_{\text{true}})$ with q' a donor question from another item; D= $(q, c_{\text{sub}}, a_{\text{true}})$ with an entity-swapped context. Cell B = $(q, \text{paraphrase}(c_{\text{org}}), a_{\text{true}})$ would be supported but answer-absent; it is produced only when a paraphraser is supplied and an item passes a *dual* filter: token-overlap of the answer with the rewrite ≤ 0.34 *and* NLI entailment of the answer by the rewrite ≥ 0.5. Greedy rewrites that remove the answer string tend to fall below the entailment threshold, while those that stay entailed tend to retain the answer tokens; the released build supplies no paraphraser, so cell B has zero yield and experiments use cells A/C/D.

RAGTruth labels. Each (context, response) row gets one binary response-level label: FAITHFUL$=1$ iff the dataset's `hallucination_labels_processed` dictionary sums to zero (no flagged span). We use no sentence-level mapping; the sentence-level SummaC baseline (§5) decomposes the response at *scoring* time only. The same 2,000 fixed responses are scored by every probe model (hidden states/confidence extracted by teacher-forcing each response through that model), which is why the NLI and lexical-overlap columns are identical across models. The whole-response NLI baseline forms its hypothesis by declarativizing each (question, answer) pair via a fixed template and scores the entailment probability with the context as premise; the sentence-level variant instead uses each response sentence as a hypothesis (§5).

Probing and cross-validation. Directions are mass-mean (difference of class means). Within-stratum decodability uses 5-fold grouped cross-validation with the groups defined by *source*, not by record. For CtrlPairs the group is the normalized (question, context, answer) triple of the underlying NQ-Swap QA: ITEM_ID indexes one swap record, and because several records share a source QA the 1,200 cell-A rows cover only 555 distinct triples, so grouping on ITEM_ID left 903 rows with a byte-identical twin in another fold. For the in-domain RAGTruth probe the group is the normalized source passage: the 2,000 released responses come from 450 passages (4–5 each), while ITEM_ID is a per-response counter, so the earlier grouping was row-level. Confidence and surface purging is fitted *within* each training fold and applied to that fold's test rows. Cross-family transfer fits an orthogonal Procrustes map on shared stimuli at each model's a-priori layer (nearest cached layer to relative depth 0.5), with the calibration/evaluation split also grouped by source.

Evaluation rigor. Moving from ITEM_ID groups to source-QA groups changes the C1 estimates by at most 0.007 (raw by $\leq$ 0.0004), and moving the RAGTruth probe from response groups to passage groups changes it by at most 0.008; both leaks were real but numerically inert, because a difference of class means in $\sim$3500 dimensions is shifted only $O(1/n)$ by a duplicated row. We resolve axis polarity on the train (or calibration) fold only and score the test projection as-is: the test-fold $\max(\mathrm{AUC}, 1-\mathrm{AUC})$ convention floors a non-predictive direction above 0.5 (it inflates a true-chance random direction to $\approx$ 0.54–0.59), so we use it nowhere. The C1 analysis layer is fixed a priori at relative depth 0.5 (as C3/C4 already do) rather than argmax-selected on the reported metric. Every reported number is produced by a single script, `scripts/cr_final_tables.py`, under this one protocol; `MANIFEST.md` maps each table, figure and quoted value to its producer and inputs.

Compute. All activation extraction runs on a single A100-40G; per-model batch sizes are reduced for large-vocabulary models to avoid OOM (Qwen 4, Gemma 2, Llama 6, Mistral 16). Parameter counts are 7.6B (Qwen2.5-7B-Instruct), 7.2B (Mistral-7B-Instruct-v0.3), 8.0B (Llama-3.1-8B-Instruct) and 9.2B (Gemma-2-9B-it); no model is trained or fine-tuned, so the budget is inference only. It totals 58,716 forward passes: per family, 3,593 CtrlPairs statements, 2,400 closed-book parametric-belief scorings (1,200 items $\times$ two candidate answers) and 3,593 each for the no-context and shuffled-context ablations, plus 2,000 RAGTruth responses for three families. The whole campaign ran on one A100-40G within a single day; our extraction logs do not instrument per-run wall-clock, so we report the forward-pass budget rather than a GPU-hour figure. All analyses, the sentence-level NLI baseline, and figures run on CPU/Apple-MPS without a GPU. The Mistral RAGTruth extraction was the one run that did not complete: it reuses the batch size tuned for the shorter CtrlPairs prompts (16) at the longer RAGTruth context length (2048 tokens) and exhausts memory, which is why C4 covers 3/4 families. Re-running it at the batch size used for Qwen would fix this; we report the negative on the three families we extracted.